MIL-T-458A
28 April 1969
SUPERSEDING
JAN-T-458
7 March 1947

MILITARY SPECIFICATION

TIN, PULVERIZED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers pulverized tin for use in smokeless powder.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves, Test

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6810

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIAL

ASTM D1193-66 - Specification for Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Material. The pulverized tin shall be a product of high quality, suitable for the purpose intended, and so formulated as to meet the requirements specified herein.

3.2 Purity. The pulverized tin shall be 99 percent pure, minimum, when tested as specified in 4.5.2.

3.3 Granulation. The pulverized tin shall conform to the granulation of table I when tested as specified in 4.5.3.

Table I. Granulation requirements

Through Sieve No.	Percent, minimum
80	100.0
100	99.5
200	98.0
230	95.0
325	93.0

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Lot size. An inspection lot shall consist of not more than 10,000 pounds of pulverized tin. If the material is produced by a continuous-run process, the lot shall contain material from only one continuous-run. Material in the inspection lot shall be identified by order or production (in the case of a continuous-run process), or by a batch number (in the case of a batch process) until final action is taken on the acceptance or rejection of the lot.

4.3 Sampling. The inspector shall take three separate 1 pound samples from the lot. If the material was produced in batches, each sample shall be taken from a different batch, if possible. If the material was produced in one continuous run, the three samples shall be taken as follows: one from the first part; one from the middle part; and one from the last part of the run. Each sample shall be placed in a clean, dry, metal or glass container which shall be sealed and carefully marked.

4.4 Lot acceptance. Each of the three samples as selected in 4.3 shall be subjected to all the tests specified in 4.5. If any sample is found to be not in conformance with the requirements of this specification, the lot represented shall be rejected.

4.5 Test methods.

4.5.1 General. Distilled water, in accordance with ASTM D1193 and analytical grade reagents shall be used throughout the tests. Blank determinations shall be run in parallel with the tests, using the same quantities of reagents used in the tests, and corrections shall be applied when necessary.

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4.5.2 Purity. Transfer a weighed portion of approximately 0.25 gm. to a 500 ml. flask. Dissolve in 50 ml. of concentrated HCl. When solution is complete, wash down the walls and dilute to 100 ml. Add 75 ml. of concentrated HCl, 25 ml. of a 40 percent H_2SO solution, 100 ml. water and 3 gm. pure lead. Close the flask with a two-hole rubber stopper, one hole of which carries a gas-inlet tube and the other a reflux condenser. Connect the inlet tube to a source of carbon dioxide gas, start a slow current of gas through the flask, slowly heat the contents of the flask to boiling, and continue boiling for 30 minutes. If the carbon dioxide is drawn from a pressure tank, provide a safety valve by inserting a T-tube in the gas line and submerging the open end in a test tube containing mercury. At the end of the boiling period, remove the stopper from the flask and insert a three-hole stopper carrying: (1) a gas-inlet tube connected to the source of the carbon dioxide used; (2) a gas-exit tube extending through a two-hole stopper to the bottom of a second flask containing a 10 percent sodium carbonate solution; and (3) a glass plug. Increase the rate of flow of carbon dioxide and cease heating the flask. Cool the flask and contents to about 10°C by means of an ice bath. Be sure that air is excluded by keeping the end of the gas-exit tube below the surface of the sodium-carbonate solution. When cooling is complete, remove the glass plug and the gas-exit tube, but continue the current of carbon dioxide. By means of a pipet, add 5 ml. of clear starch solution, and then insert the tip of a buret containing 0.1N iodine solution. Titrate to a permanent blue color. The iodine solution shall have been previously standardized against pure tin. Correct the result by means of a blank determination run on the reagents. Calculate the percent purity as follows:

$$\text{Percent purity} = \frac{5.935\ (V-v)\ N}{W}$$

where: V = ml. iodine solution required in titration
v = ml. iodine solution required in blank
N = normality of iodine solution
W = gm. sample

4.5.3 Granulation. Place a weighed portion of approximately 50 gm. of the sample on the next of sieves specified in 3.3, properly superimposed and assembled with a cover and a bottom pan. The sieves shall conform to RR-S-366. Shake for 30 minutes by means of a mechanically operated sieve shaker which imparts to the sieves a rotary motion and tapping action of uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the material remaining on each sieve and calculate the percent passing through each sieve.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Pulverized tin shall be packed in steel kegs, steel drums or steel pails of the type, size and kind commonly used for the purpose and so constructed as to insure acceptance and safe delivery by common or other carrier, at the lowest rate, to the point of delivery.

5.2 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Packing required (see 5.1)
(c) Unit quantity required.

6.2 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

Custodian:
Army - MU
Navy - OS

Review activities:
Army - MU, MD, MI
Navy - OS

Preparing activity:
Army - MU

Project No. 6810-0485

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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